

otherwise. The first case only happens in the most severe case of sparsity: when a gradient has been zero at all timesteps except at the current timestep. For less sparse cases, the effective stepsize will be smaller. When $(1 - \beta_1) = \sqrt{1 - \beta_2}$ we have that $|\hat{m}_t/\sqrt{\hat{v}_t}| < 1$ therefore $|\Delta_t| < \alpha$. In more common scenarios, we will have that $\hat{m}_t/\sqrt{\hat{v}_t} \approx \pm 1$ since $|\mathbb{E}[g]/\sqrt{\mathbb{E}[g^2]}| \leq 1$. The effective magnitude of the steps taken in parameter space at each timestep are approximately bounded by the stepsize setting α , i.e., $|\Delta_t| \lesssim \alpha$. This can be understood as establishing a *trust region* around the current parameter value, beyond which the current gradient estimate does not provide sufficient information. This typically makes it relatively easy to know the right scale of α in advance. For many machine learning models, for instance, we often know in advance that good optima are with high probability within some set region in parameter space; it is not uncommon, for example, to have a prior distribution over the parameters. Since α sets (an upper bound of) the magnitude of steps in parameter space, we can often deduce the right order of magnitude of α such that optima can be reached from θ_0 within some number of iterations. With a slight abuse of terminology, we will call the ratio $\hat{m}_t/\sqrt{\hat{v}_t}$ the *signal-to-noise ratio (SNR)*. With a smaller SNR the effective stepsize Δ_t will be closer to zero. This is a desirable property, since a smaller SNR means that there is greater uncertainty about whether the direction of $\hat{m}_t$ corresponds to the direction of the true gradient. For example, the SNR value typically becomes closer to 0 towards an optimum, leading to smaller effective steps in parameter space: a form of automatic annealing. The effective stepsize Δ_t is also invariant to the scale of the gradients; rescaling the gradients g with factor c will scale $\hat{m}_t$ with a factor c and $\hat{v}_t$ with a factor c^2 , which cancel out: $(c \cdot \hat{m}_t)/(\sqrt{c^2 \cdot \hat{v}_t}) = \hat{m}_t/\sqrt{\hat{v}_t}$.

3 INITIALIZATION BIAS CORRECTION

As explained in section 2, Adam utilizes initialization bias correction terms. We will here derive the term for the second moment estimate; the derivation for the first moment estimate is completely analogous. Let g be the gradient of the stochastic objective f , and we wish to estimate its second raw moment (uncentered variance) using an exponential moving average of the squared gradient, with decay rate β_2 . Let $g_1, \dots, g_T$ be the gradients at subsequent timesteps, each a draw from an underlying gradient distribution $g_t \sim p(g_t)$. Let us initialize the exponential moving average as $v_0 = 0$ (a vector of zeros). First note that the update at timestep t of the exponential moving average $v_t = \beta_2 \cdot v_{t-1} + (1 - \beta_2) \cdot g_t^2$ (where g_t^2 indicates the elementwise square $g_t \odot g_t$) can be written as a function of the gradients at all previous timesteps:

$$v_t = (1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} \cdot g_i^2 \quad (1)$$

We wish to know how $\mathbb{E}[v_t]$, the expected value of the exponential moving average at timestep t , relates to the true second moment $\mathbb{E}[g_t^2]$, so we can correct for the discrepancy between the two. Taking expectations of the left-hand and right-hand sides of eq. (1):

$$\mathbb{E}[v_t] = \mathbb{E} \left[(1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} \cdot g_i^2 \right] \quad (2)$$

$$= \mathbb{E}[g_t^2] \cdot (1 - \beta_2) \sum_{i=1}^t \beta_2^{t-i} + \zeta \quad (3)$$

$$= \mathbb{E}[g_t^2] \cdot (1 - \beta_2^t) + \zeta \quad (4)$$

where $\zeta = 0$ if the true second moment $\mathbb{E}[g_t^2]$ is stationary; otherwise ζ can be kept small since the exponential decay rate β_1 can (and should) be chosen such that the exponential moving average assigns small weights to gradients too far in the past. What is left is the term $(1 - \beta_2^t)$ which is caused by initializing the running average with zeros. In algorithm 1 we therefore divide by this term to correct the initialization bias.

In case of sparse gradients, for a reliable estimate of the second moment one needs to average over many gradients by choosing a small value of β_2 ; however it is exactly this case of small β_2 where a lack of initialisation bias correction would lead to initial steps that are much larger.

Algorithm 2: *AdaMax*, a variant of Adam based on the infinity norm. See section 7.1 for details. Good default settings for the tested machine learning problems are $\alpha = 0.002$, $\beta_1 = 0.9$ and $\beta_2 = 0.999$. With β_1^t we denote β_1 to the power t . Here, $(\alpha/(1 - \beta_1^t))$ is the learning rate with the bias-correction term for the first moment. All operations on vectors are element-wise.

Require: α : Stepsize

Require: $\beta_1, \beta_2 \in [0, 1]$: Exponential decay rates

Require: $f(\theta)$: Stochastic objective function with parameters θ

Require: θ_0 : Initial parameter vector

$m_0 \leftarrow 0$ (Initialize 1st moment vector)

$u_0 \leftarrow 0$ (Initialize the exponentially weighted infinity norm)

$t \leftarrow 0$ (Initialize timestep)

while θ_t not converged **do**

$t \leftarrow t + 1$

$g_t \leftarrow \nabla_{\theta} f_t(\theta_{t-1})$ (Get gradients w.r.t. stochastic objective at timestep t)

$m_t \leftarrow \beta_1 \cdot m_{t-1} + (1 - \beta_1) \cdot g_t$ (Update biased first moment estimate)

$u_t \leftarrow \max(\beta_2 \cdot u_{t-1}, |g_t|)$ (Update the exponentially weighted infinity norm)

$\theta_t \leftarrow \theta_{t-1} - (\alpha/(1 - \beta_1^t)) \cdot m_t/u_t$ (Update parameters)

end while

return θ_t (Resulting parameters)

Note that the decay term is here equivalently parameterised as β_2^p instead of β_2 . Now let $p \rightarrow \infty$, and define $u_t = \lim_{p \rightarrow \infty} (v_t)^{1/p}$, then:

$$u_t = \lim_{p \rightarrow \infty} (v_t)^{1/p} = \lim_{p \rightarrow \infty} \left((1 - \beta_2^p) \sum_{i=1}^t \beta_2^{p(t-i)} \cdot |g_i|^p \right)^{1/p} \quad (8)$$

$$= \lim_{p \rightarrow \infty} (1 - \beta_2^p)^{1/p} \left(\sum_{i=1}^t \beta_2^{p(t-i)} \cdot |g_i|^p \right)^{1/p} \quad (9)$$

$$= \lim_{p \rightarrow \infty} \left(\sum_{i=1}^t \left(\beta_2^{(t-i)} \cdot |g_i| \right)^p \right)^{1/p} \quad (10)$$

$$= \max(\beta_2^{t-1}|g_1|, \beta_2^{t-2}|g_2|, \dots, \beta_2|g_{t-1}|, |g_t|) \quad (11)$$

Which corresponds to the remarkably simple recursive formula:

$$u_t = \max(\beta_2 \cdot u_{t-1}, |g_t|) \quad (12)$$

with initial value $u_0 = 0$. Note that, conveniently enough, we don't need to correct for initialization bias in this case. Also note that the magnitude of parameter updates has a simpler bound with AdaMax than Adam, namely: $|\Delta_t| \leq \alpha$.

7.2 TEMPORAL AVERAGING

Since the last iterate is noisy due to stochastic approximation, better generalization performance is often achieved by averaging. Previously in Moulines & Bach (2011), Polyak-Ruppert averaging (Polyak & Juditsky, 1992; Ruppert, 1988) has been shown to improve the convergence of standard SGD, where $\bar{\theta}_t = \frac{1}{t} \sum_{k=1}^t \theta_k$. Alternatively, an exponential moving average over the parameters can be used, giving higher weight to more recent parameter values. This can be trivially implemented by adding one line to the inner loop of algorithms 1 and 2: $\bar{\theta}_t \leftarrow \beta_2 \cdot \bar{\theta}_{t-1} + (1 - \beta_2)\theta_t$, with $\bar{\theta}_0 = 0$. Initialization bias can again be corrected by the estimator $\hat{\theta}_t = \bar{\theta}_t/(1 - \beta_2^t)$.

8 CONCLUSION

We have introduced a simple and computationally efficient algorithm for gradient-based optimization of stochastic objective functions. Our method is aimed towards machine learning problems with

10 APPENDIX

10.1 CONVERGENCE PROOF

Definition 10.1. A function $f : R^d \rightarrow R$ is convex if for all $x, y \in R^d$, for all $\lambda \in [0, 1]$,

$$\lambda f(x) + (1 - \lambda)f(y) \geq f(\lambda x + (1 - \lambda)y)$$

Also, notice that a convex function can be lower bounded by a hyperplane at its tangent.

Lemma 10.2. If a function $f : R^d \rightarrow R$ is convex, then for all $x, y \in R^d$,

$$f(y) \geq f(x) + \nabla f(x)^T (y - x)$$

The above lemma can be used to upper bound the regret and our proof for the main theorem is constructed by substituting the hyperplane with the Adam update rules.

The following two lemmas are used to support our main theorem. We also use some definitions simplify our notation, where $g_t \triangleq \nabla f_t(\theta_t)$ and $g_{t,i}$ as the i^{th} element. We define $g_{1:t,i} \in \mathbb{R}^t$ as a vector that contains the i^{th} dimension of the gradients over all iterations till t , $g_{1:t,i} = [g_{1,i}, g_{2,i}, \dots, g_{t,i}]$

Lemma 10.3. Let $g_t = \nabla f_t(\theta_t)$ and $g_{1:t}$ be defined as above and bounded, $\|g_t\|_2 \leq G$, $\|g_t\|_\infty \leq G_\infty$. Then,

$$\sum_{t=1}^T \sqrt{\frac{g_{t,i}^2}{t}} \leq 2G_\infty \|g_{1:T,i}\|_2$$

Proof. We will prove the inequality using induction over T .

The base case for $T = 1$, we have $\sqrt{g_{1,i}^2} \leq 2G_\infty \|g_{1,i}\|_2$.

For the inductive step,

$$\begin{aligned} \sum_{t=1}^T \sqrt{\frac{g_{t,i}^2}{t}} &= \sum_{t=1}^{T-1} \sqrt{\frac{g_{t,i}^2}{t}} + \sqrt{\frac{g_{T,i}^2}{T}} \\ &\leq 2G_\infty \|g_{1:T-1,i}\|_2 + \sqrt{\frac{g_{T,i}^2}{T}} \\ &= 2G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} + \sqrt{\frac{g_{T,i}^2}{T}} \end{aligned}$$

From, $\|g_{1:T,i}\|_2^2 - g_{T,i}^2 + \frac{g_{T,i}^4}{4\|g_{1:T,i}\|_2^2} \geq \|g_{1:T,i}\|_2^2 - g_{T,i}^2$, we can take square root of both side and have,

$$\begin{aligned} \sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} &\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\|g_{1:T,i}\|_2} \\ &\leq \|g_{1:T,i}\|_2 - \frac{g_{T,i}^2}{2\sqrt{T}G_\infty} \end{aligned}$$

Rearrange the inequality and substitute the $\sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2}$ term,

$$G_\infty \sqrt{\|g_{1:T,i}\|_2^2 - g_{T,i}^2} + \sqrt{\frac{g_{T,i}^2}{T}} \leq 2G_\infty \|g_{1:T,i}\|_2$$

□

Lemma 10.4. Let $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. For $\beta_1, \beta_2 \in [0, 1)$ that satisfy $\frac{\beta_1^2}{\sqrt{\beta_2}} < 1$ and bounded g_t , $\|g_t\|_2 \leq G$, $\|g_t\|_\infty \leq G_\infty$, the following inequality holds

$$\sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} \leq \frac{2}{1-\gamma} \frac{1}{\sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

Proof. Under the assumption, $\frac{\sqrt{1-\beta_2^t}}{(1-\beta_1^t)^2} \leq \frac{1}{(1-\beta_1)^2}$. We can expand the last term in the summation using the update rules in Algorithm 1,

$$\begin{aligned} \sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} &= \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(\sum_{k=1}^T (1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T \sum_{j=1}^T (1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^T \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T \sum_{j=1}^T (1-\beta_2)\beta_2^{T-j} g_{j,i}^2}} \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \sum_{k=1}^T \frac{T((1-\beta_1)\beta_1^{T-k} g_{k,i})^2}{\sqrt{T(1-\beta_2)\beta_2^{T-k} g_{k,i}^2}} \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{\sqrt{1-\beta_2^T}}{(1-\beta_1^T)^2} \frac{(1-\beta_1)^2}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^T T \left(\frac{\beta_1^2}{\sqrt{\beta_2}} \right)^{T-k} \|g_{k,i}\|_2 \\ &\leq \sum_{t=1}^{T-1} \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} + \frac{T}{\sqrt{T(1-\beta_2)}} \sum_{k=1}^T \gamma^{T-k} \|g_{k,i}\|_2 \end{aligned}$$

Similarly, we can upper bound the rest of the terms in the summation.

$$\begin{aligned} \sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} &\leq \sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^{T-t} t\gamma^j \\ &\leq \sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^T t\gamma^j \end{aligned}$$

For $\gamma < 1$, using the upper bound on the arithmetic-geometric series, $\sum_t t\gamma^t < \frac{1}{(1-\gamma)^2}$:

$$\sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t(1-\beta_2)}} \sum_{j=0}^T t\gamma^j \leq \frac{1}{(1-\gamma)^2 \sqrt{1-\beta_2}} \sum_{t=1}^T \frac{\|g_{t,i}\|_2}{\sqrt{t}}$$

Apply Lemma 10.3,

$$\sum_{t=1}^T \frac{\hat{m}_{t,i}^2}{\sqrt{t\hat{v}_{t,i}}} \leq \frac{2G_\infty}{(1-\gamma)^2 \sqrt{1-\beta_2}} \|g_{1:T,i}\|_2$$

□

To simplify the notation, we define $\gamma \triangleq \frac{\beta_1^2}{\sqrt{\beta_2}}$. Intuitively, our following theorem holds when the learning rate α_t is decaying at a rate of $t^{-\frac{1}{2}}$ and first moment running average coefficient $\beta_{1,t}$ decay exponentially with λ , that is typically close to 1, e.g. $1 - 10^{-8}$.

Theorem 10.5. Assume that the function f_t has bounded gradients, $\|\nabla f_t(\theta)\|_2 \leq G$, $\|\nabla f_t(\theta)\|_\infty \leq G_\infty$ for all $\theta \in \mathbb{R}^d$ and distance between any θ_t generated by Adam is bounded, $\|\theta_n - \theta_m\|_2 \leq D$,